

TRACK 4 — OPSGENIE AI

OpsGenie AI

An AI operations control tower for cold-chain logistics: predictive maintenance, spoilage risk, demand forecasting and gated workflow automation in one causal pipeline

A dashboard shows you metrics. A control tower tells you what to do next.

Team 40

TEAM MEMBERS

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0.987

ROC-AUC, reefer failure within 24 h (held-out units, observable windows)

58%

lower demand-forecast error than a 7-day moving average

₹1.92 L

loss avoided by one approved reroute on a ₹5 L frozen load

₹1L

ceiling above which no action runs without a human approval

Working prototype

Live application	https://opsgenie-ai-piba.onrender.com/
Live API + docs	https://opsgenie-ai-piba.onrender.com/docs
Source code	https://github.com/ShubhamAdkhale/OpsGenieAI

Note for judges: weather is live from Open-Meteo across 14 lane waypoints; sensors, traffic and demand run on the app's own digital twin and every value on screen is labelled LIVE, SIMULATED or PREDICTED. Press **Reset** before a run, or click ► **Guided demo** to replay the whole scenario on its own.

2 Problem Statement

A cold-chain operator does not lose money because one tool fails. It loses money because its tools do not talk to each other. A refrigeration unit starts to wear, a lane slows down in traffic and a heat wave rolls in. Each event shows up on a different screen, and nobody joins them up before a ₹5 lakh load of frozen chicken arrives spoiled.

Core problem. Build one system that links **equipment health, transit conditions, spoilage risk and inventory** into a single decision that an operations manager can act on: what is going wrong, what will happen, why, how much money is at stake and what to do. Low-risk actions should run on their own; high-impact actions should wait for a human.

3 Problem Understanding

Background: how a cold-chain load is actually lost

Frozen food must stay at -18 °C, with brief rises to no warmer than -15 °C in transit (the Codex quick-frozen rule). A reefer truck holds that line only while its cooling power beats the heat leaking in from outside. A refrigerant leak cuts the cooling. A hot day raises the heat ingress. A traffic jam keeps the cargo on the road longer. **None of these alone spoils the load. Together they do.** Today they are watched by three different teams on three different systems: maintenance on a CMMS, transit on a GPS tracker, stock on a spreadsheet.

Who is affected

PERSONA	WHAT THEY NEED	WHAT GOES WRONG TODAY
Operations manager (control tower)	Which load needs attention now, and the best action	A wall of metrics and alerts; nothing ranked by money at risk
Maintenance engineer	Which unit will fail next, and which cargo it is carrying	Fixed service intervals; faults found after the breakdown
Supply / inventory planner	How much to reorder before a stockout	Moving-average forecasts; spoiled loads show up as surprise shortages
Business owner / finance	Lower spoilage write-offs without losing control	Either everything is manual, or automation acts with no limits

Pain points we set out to fix

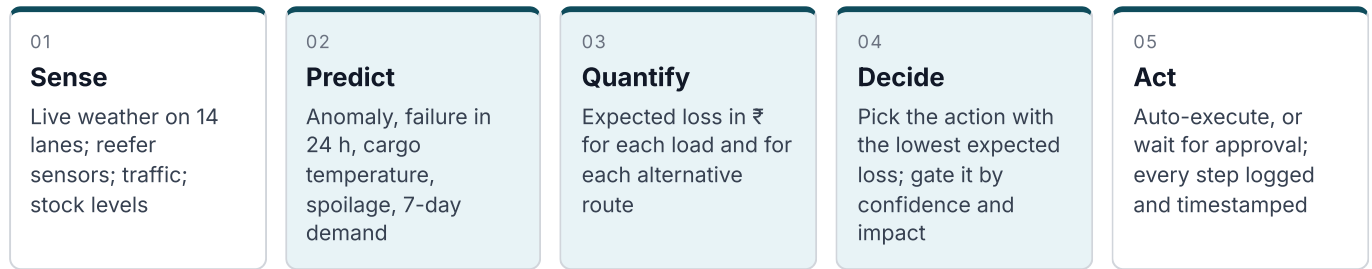
- **Siloed signals.** Maintenance does not know which shipment a failing reefer is carrying. Dispatch does not know the reefer is failing.
- **Metrics, not decisions.** "Temperature -12 °C" does not tell a manager whether to reroute, swap the unit or carry on.
- **No money attached.** A risk score of 0.7 means nothing to finance. "₹3.9 L expected loss, ₹1.9 L avoidable" does.
- **Reactive maintenance.** Units are serviced on a calendar or after they break, not when their sensor trend says so.
- **Unsafe automation.** Rules engines either act on everything or nothing. There is no gate based on confidence and impact.
- **Opaque AI.** A model that will not say why it flagged a load cannot be trusted with a ₹5 L decision.

Commercial and social impact

Spoiled perishables are a direct write-off, a missed delivery and a lost customer. The cost falls hardest on mid-sized distributors with no data team. Every load saved is food that is not wasted, and stock that reaches a pharmacy, a retailer or a restaurant on time.

4 Proposed Solution

OpsGenie AI replaces Sensor → Alert → Human guesses with a five-stage pipeline. Every asset in the network moves through it every three seconds, and each stage passes its output to the next.



Five risk bands from one transparent score

Each shipment gets a **Melt Index** (0–100): a weighted score of six risk components. It is deliberately a published formula, not a black box, so every point can be traced to a cause.

BAND	MELT INDEX	WHAT IT MEANS	WHAT THE SYSTEM DOES
SAFE	0–20	Cargo cold, margin intact	Monitor silently
LOW	21–40	One factor drifting (e.g. delay)	Watch; no action needed yet
MODERATE	41–60	Two or more factors stacking	Evaluate alternative routes; raise a warning
HIGH	61–80	Spoilage likely on the current plan	Recommend a reroute or unit swap; open a workflow
CRITICAL	81–100	Cold chain breached or about to	Top of the control tower; manager approval requested

Every recommendation is priced

Expected loss	$\text{value} \times P(\text{spoilage})$
Expected loss, per route	$\text{value} \times P(\text{spoilage} \mid \text{route, reefer})$
Loss avoided	current - best route

The optimizer minimizes **expected business loss**, not travel time. A reroute must also save at least ₹15,000 *and* 5% of the load's value, so it does not churn trucks for trivial gains.

Automation with a safety gate

IMPACT	CONFIDENCE	ACTION
> ₹1,00,000	any	Human approval, always
₹25K – ₹1L	any	Recommend, pending approval
< ₹25,000	HIGH (≥ 0.70)	Auto-execute and log
any	LOW (< 0.40)	Recommend only, flagged

One decision, answered for the people who need it

OUTPUT	READER	PURPOSE
Priority card	Operations manager	What is wrong, what will happen, why, how much is at risk (with the arithmetic) and what to do, plus an Approve button
Shipment detail	Operations analyst	Heat balance in plain words, each Melt Index component in points, route A vs B side by side
Machine health	Maintenance engineer	Anomaly score, failure-in-24 h probability, wear rate, sensor trends, and which shipment the unit is carrying
Forecast & reorder	Supply planner	7-day demand per product, stockout probability and reorder quantity with 15% safety stock
Workflow log	Owner / auditor	Every recommendation, approval, override and execution, timestamped and immutable

5 Innovation & Uniqueness

CAPABILITY	TYPICAL OPS DASHBOARD	OPSGENIE AI
Scope	Separate tools for maintenance, tracking and stock	One pipeline: a machine's health is a live input to the shipment it carries and to the stock it feeds
Output	Charts and threshold alerts	A ranked decision with probability, ₹ expected loss, causes and one recommended action
Maintenance	Calendar service, or after breakdown	IsolationForest anomaly detection plus a failure-in-24 h classifier that reads the sensor trend
Routing	Fastest route	Lowest expected loss, including handing the load to a healthy standby reefer at a cold hub
Automation	All manual, or unrestricted rules	Confidence × impact gate; ₹1 L hard ceiling for human approval; approvals immutable
Explainability	None, or a feature-importance chart	Every score broken into weighted points, plus a heat-balance sentence written from the physics
Data honesty	Demo data shown as if it were real	Every value labelled LIVE / SIMULATED / PREDICTED; a simulated reading is never shown as live

The moat: consequences are computed, never assigned

Most demo dashboards hard-code their numbers. OpsGenie integrates them. A scenario may only change a **root cause**: a traffic delay, a wear rate or an ambient temperature. Then the simulation runs, and what appears on screen is whatever the models make of the new conditions. Below is the hero shipment SC-1042 (₹5 L of frozen chicken, Mumbai → Pune) as measured on the live prototype:

#	ROOT CAUSE APPLIED	BAND	CARGO	EXCURSION	P(SPOIL)	EXPECTED LOSS	
0	baseline (after Reset)	SAFE 19	−18.0 °C	0 min	8%	₹42,000	
1	Traffic incident, +95 min delay	LOW 37	−18.3 °C	0 min	19%	₹97,000	
2	Refrigerant leak, RF-204 wears 8 pts/h	HIGH 64	−17.3 °C	0 min	53%	₹2,63,000	
3	Heat wave +11 °C, lane turns STORM	CRITICAL 84	−7.2 °C	105 min	77%	₹3,87,000	
4	Manager approves reroute via Chakan cold hub to standby reefer RF-205: expected loss ₹3.91 L → ₹1.99 L, so ₹1.92 L avoided						

Notice step 1: the traffic jam raised risk but added **zero** excursion minutes, because the cargo never warmed up. A delay does not spoil food directly; it keeps it in a truck for longer. Exact figures move a little with live weather; the shape of the escalation is enforced by tests.

Honest by design

- **Physics, not scripts.** Cargo temperature follows Newton's law of cooling; shelf life burns faster when warm (the Q10 rule).
- **Excursion is earned.** Minutes outside the safe range accrue only while the cargo is truly above its limit.
- **No injected outliers.** Sensor traces are generated from the unit's real efficiency, so the anomaly detector catches genuine drift.
- **Repeatable.** A simulated clock means the same causes always give the same state.

Rules the tests enforce

- `test_no_scenario_can_assign_a_consequence`: scenarios can only name causes.
- `test_the_traffic_incident_only_touches_delay`: a jam cannot reach past the route.
- `test_a_healthy_world_stays_calm...`: the twin does not invent a crisis.
- `test_arc_is_identical_three_times_in_a_row`: same causes, same result.
- 132 automated tests in total.

6 Solution Architecture

Models train offline once and are saved as `.joblib` files. The API loads them at startup and runs two loops: a **physics tick** every 3 s that advances the digital twin and re-scores every asset, and a **weather loop** every 10 min. The React control tower polls every 4 s.

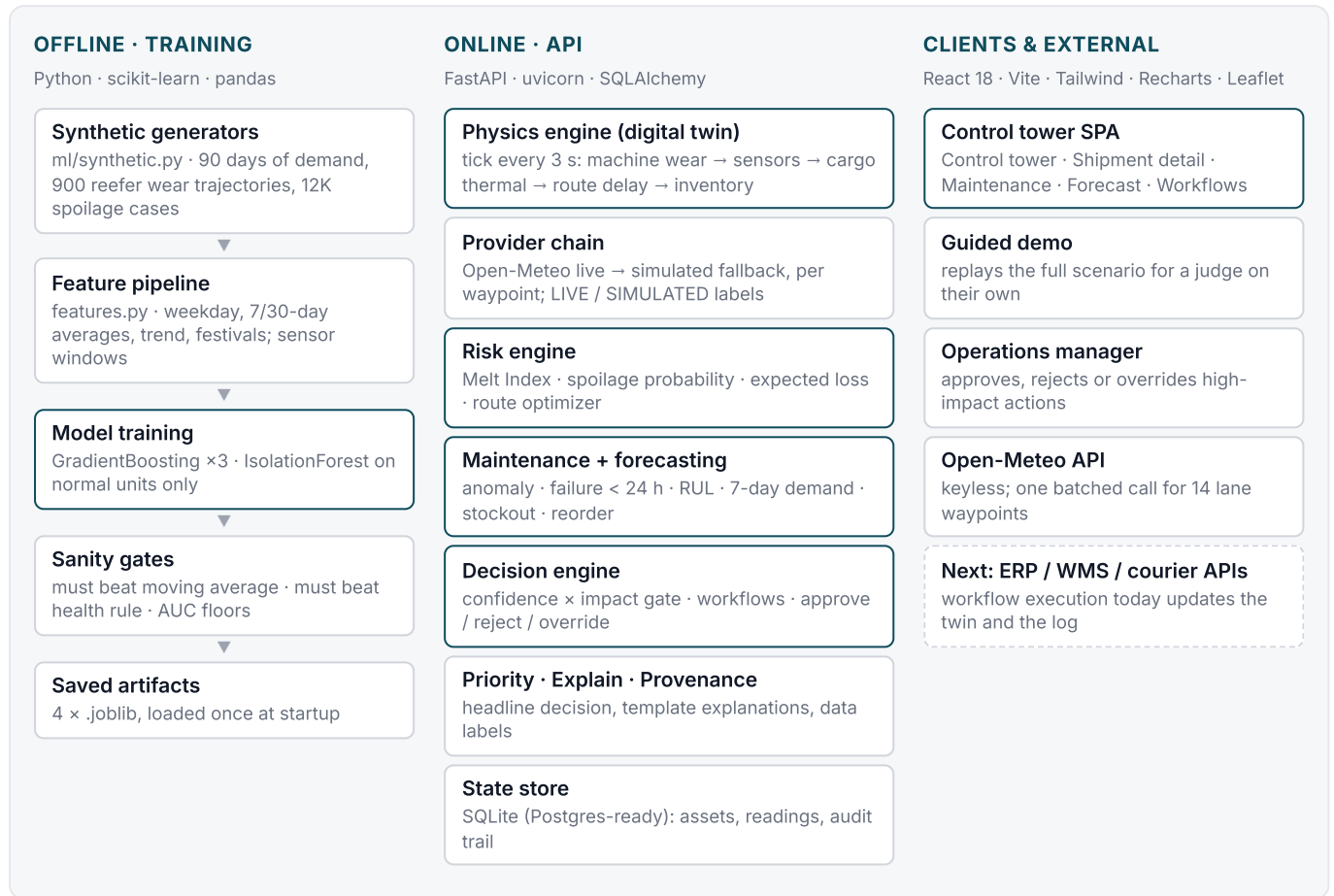


Figure 1. End-to-end architecture. Bold-outlined boxes are the core components. Every value funnels through `physics.tick()` → `risk_engine.recalculate()`, so a scenario button cannot produce a state the live simulation could not reach on its own.

Technology stack

LAYER	TECHNOLOGY	WHY THIS CHOICE
Machine learning	scikit-learn 1.6 (GradientBoosting, IsolationForest), pandas 2.2, NumPy 2.2	Fast on tabular data, CPU only, trains in about a minute
API	FastAPI 0.115 + uvicorn, Pydantic 2, SQLAlchemy 2, SQLite (Postgres-ready)	Typed JSON API with auto-generated docs at <code>/docs</code>
Frontend	React 18, Vite 6, Tailwind CSS, Recharts, React-Leaflet	Single-page control tower; static build for any CDN
Live data	Open-Meteo weather API via <code>httpx</code>	Keyless, free; one batched request for all waypoints
Quality	pytest: 132 tests across 11 files	Asserts properties (monotonic escalation, loss = value × p), not fixed numbers

Reliability controls

No outside service can break the demo. If Open-Meteo is down, simulated weather takes over and is **labelled SIMULATED**. If a model file is missing, a rule-based fallback runs and `/api/health` says so. A failing tick is logged and skipped, never left half-done.

7 AI / Technology Used

Data

Live: Open-Meteo current conditions (temperature, humidity, precipitation, wind, WMO weather code) for 14 lane waypoints across 9 Indian cities. **Simulated by the digital twin:** 12 monitored machines (9 reefers, 2 conveyors, 1 compressor), 8 active refrigerated loads, 4 products and their warehouses, sensor telemetry and traffic. **Training data:** synthetic generators, stated plainly as such. The pipeline is data-agnostic and is meant to be retrained on a real fleet's history.

The physics behind the spoilage risk

```
T_equilibrium = T_ambient - (19.2 / 0.30) × reefer_efficiency
shelf_life_burn = 2 ^ (excess °C / 10), capped at 6×
route_delay      = incident + congestion(rush hour) + weather(band)
```

Newton's law of cooling
Q10 rule: warm food ages faster
recomputed every tick

A healthy reefer holds -15°C in any weather; a worn one holds until real heat arrives; **wear plus heat together** breaks the chain. The crossover comes from the ratio of two physical constants, not from tuning the demo.

The Melt Index: a transparent risk formula

```
Melt Index = 0.25·Temperature + 0.20·Refrigeration + 0.15·Traffic
              + 0.15·Weather      + 0.15·ShelfLife      + 0.10·ETA margin
```

Each component is scored 0–100 from its own physical input (excursion minutes, 100 – reefer health, delay, weather band, shelf life used, ETA margin), and is shown on screen as points. A calibrated logistic curve maps the index to a spoilage probability; the trained classifier is one setting away.

Four trained models

MODEL	ALGORITHM	USED FOR	RESULT ON HELD-OUT DATA
Demand forecast	GradientBoostingRegressor, one per product	7-day demand, stockout risk, reorder qty	MAE 55–63% lower than a 7-day moving average on every product
Anomaly detection	IsolationForest, trained on normal units only	Anomaly score on each sensor window	0.4% false alarms; catches 98.9% of units losing 1–3 pts/h and 100% at 3–10 pts/h
Failure < 24 h	GradientBoostingClassifier on 58.5K windows	Maintenance priority, workflows	ROC-AUC 0.900 overall, 0.987 on observable windows
Spoilage	GradientBoostingClassifier on 12K loads	Optional spoilage source	98.2% of the achievable AUC (0.674 vs 0.686 label-noise ceiling)

Failure classifier vs. a health-score rule

HELD-OUT UNITS	CLASSIFIER	RULE
ROC-AUC, all windows	0.900	0.845
Brier score (lower is better)	0.076	0.247
ROC-AUC, observable windows	0.987	0.927

Evaluated on held-out *units*, not windows. "Observable" excludes the 21.8% of faults that begin after the window ends, which no sensor model can foresee. On the live demo it tracks RF-204 at 3% → 91% (leak) → 99% (heat) → 1% (repair).

Demand forecast error (MAE, units/day)

PRODUCT	MODEL	7-DAY AVG	GAIN
Frozen Chicken	18.9	44.9	–58%
Paneer	13.0	29.2	–55%
Frozen Peas	24.8	55.0	–55%
Ice Cream Tubs	20.6	55.1	–63%

Features: weekday pattern, 7- and 30-day averages, trend and festival days. Forecasts run recursively over a 7-day horizon.

No language model computes a number. Explanations are templates filled from stored assessments, so every sentence on screen can be traced back to a row in the database.

8 Prototype

The prototype is built and working end to end. All screenshots below were captured from the running application on 5 Oct 2026, driving the hero scenario live with real Open-Meteo weather.

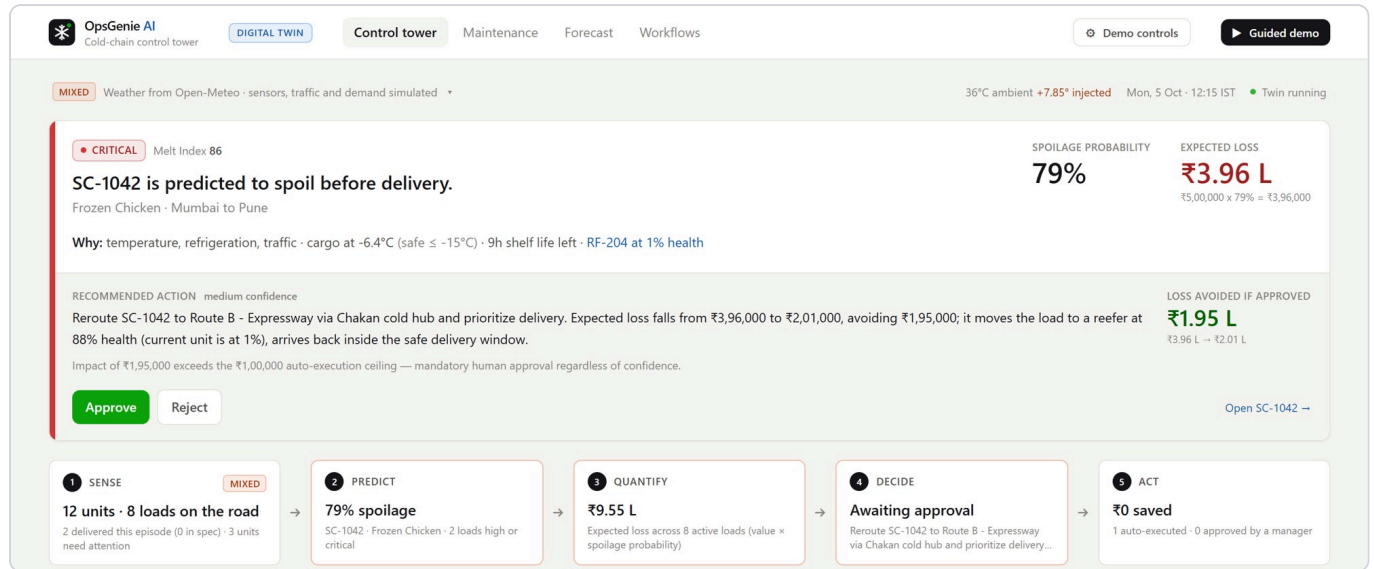


Figure 2. Control tower at the climax. The priority card answers five questions: SC-1042 is CRITICAL (Melt Index 86), 79% likely to spoil, ₹5,00,000 × 79% = ₹3.96 L expected loss, caused by temperature, refrigeration and traffic with RF-204 at 1% health. The recommended reroute avoids ₹1.95 L and, being above the ₹1 L ceiling, waits for Approve. The SENSE → ACT strip below shows the whole pipeline.

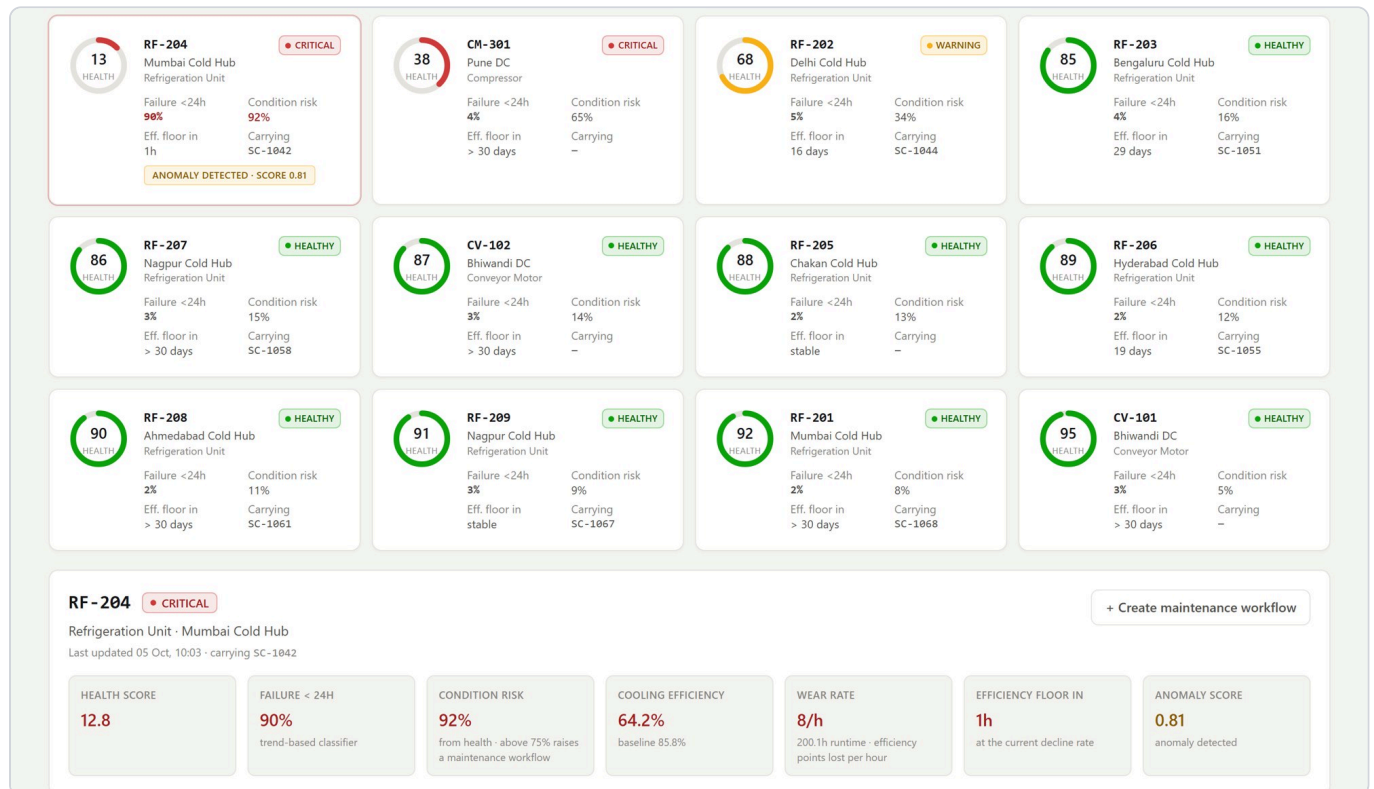


Figure 3. Predictive maintenance. Twelve monitored units ranked by health. RF-204 has fallen to 13 with an anomaly score of 0.81, a 90% chance of failure within 24 h and the efficiency floor 1 hour away, and it is carrying SC-1042. CM-301 is old and worn (65% condition risk) but declining slowly, so its 24-hour failure risk stays at 4%: the classifier reads the trend, not just the level.

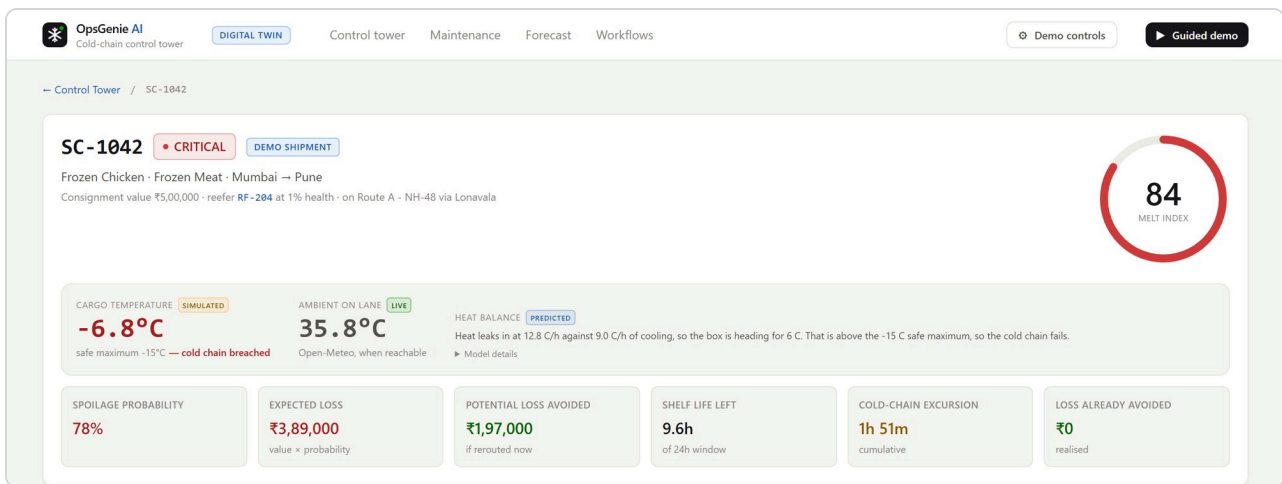


Figure 4. Shipment detail: the physics in plain words. Cargo at -6.8°C against a -15°C limit. "Heat leaks in at 12.8°C/h against 9.0°C/h of cooling, so the box is heading for 6°C ... the cold chain fails." Every value carries its source label: cargo SIMULATED, ambient LIVE, heat balance PREDICTED.

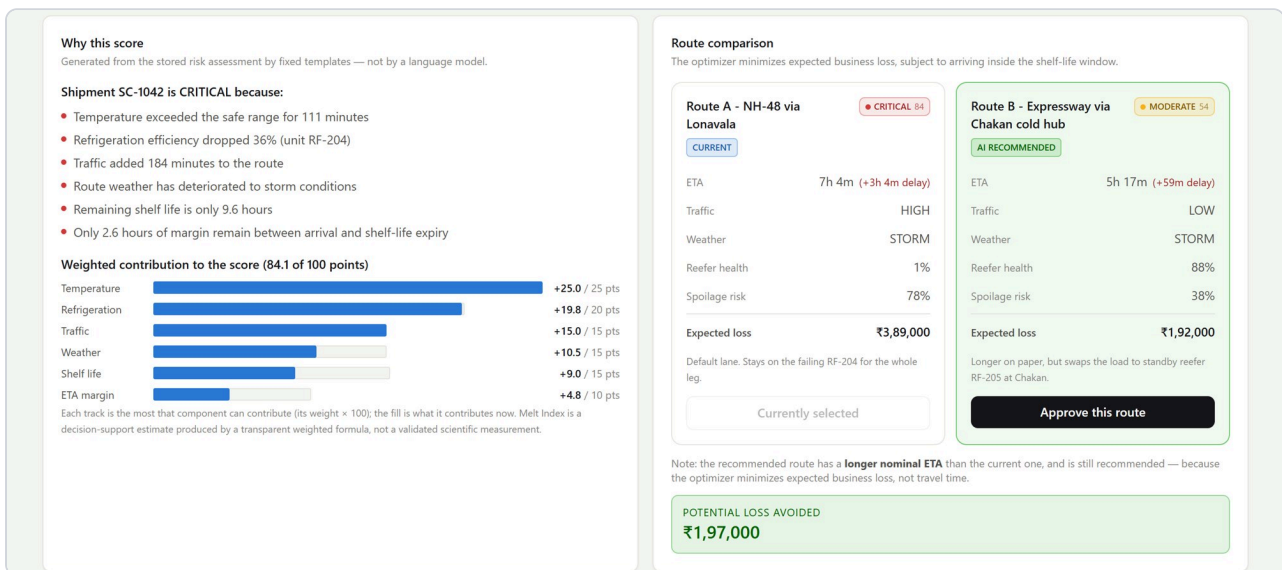


Figure 5. Why this score, and the route decision. Left: the Melt Index broken into weighted points per component. Right: Route B is longer on paper but hands the load to standby reefer RF-205 (88% health) at the Chakan cold hub, cutting spoilage risk from 78% to 38% and expected loss from ₹3.89 L to ₹1.92 L.

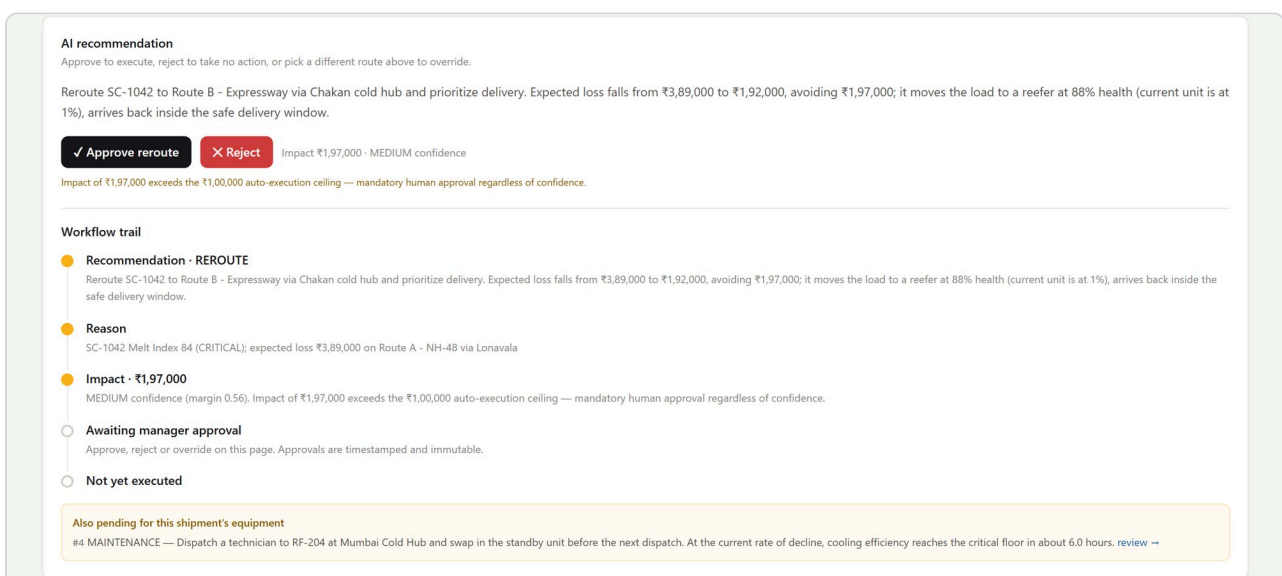


Figure 6. The AI recommendation and its workflow trail. Impact ₹1.97 L at MEDIUM confidence exceeds the auto-execution ceiling, so the manager must approve. The linked maintenance workflow for RF-204 is shown on the same page.

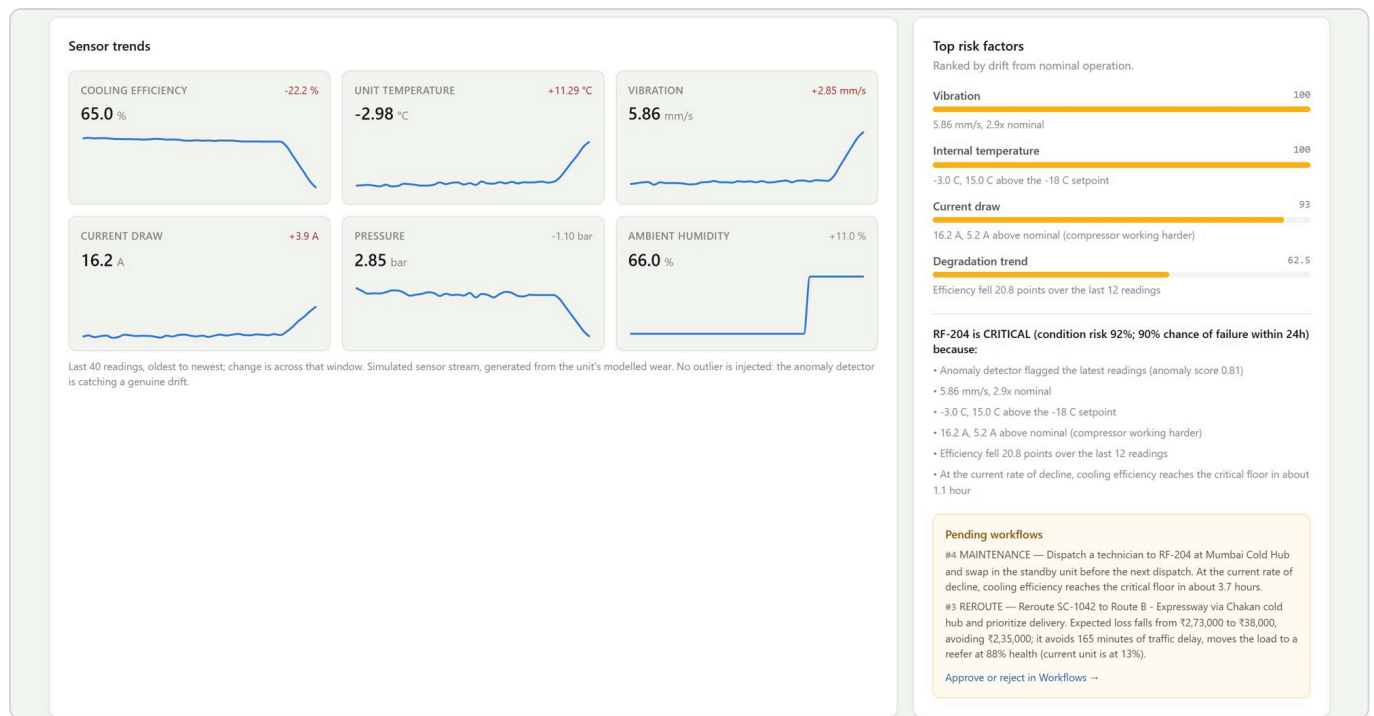


Figure 7. Sensor trends and risk factors for RF-204. Cooling efficiency, unit temperature, vibration and current draw drift together because they are all generated from the unit's modelled wear. Nothing is injected; the anomaly detector catches a genuine trajectory. The right panel ranks the drivers and lists the pending maintenance and reroute workflows.

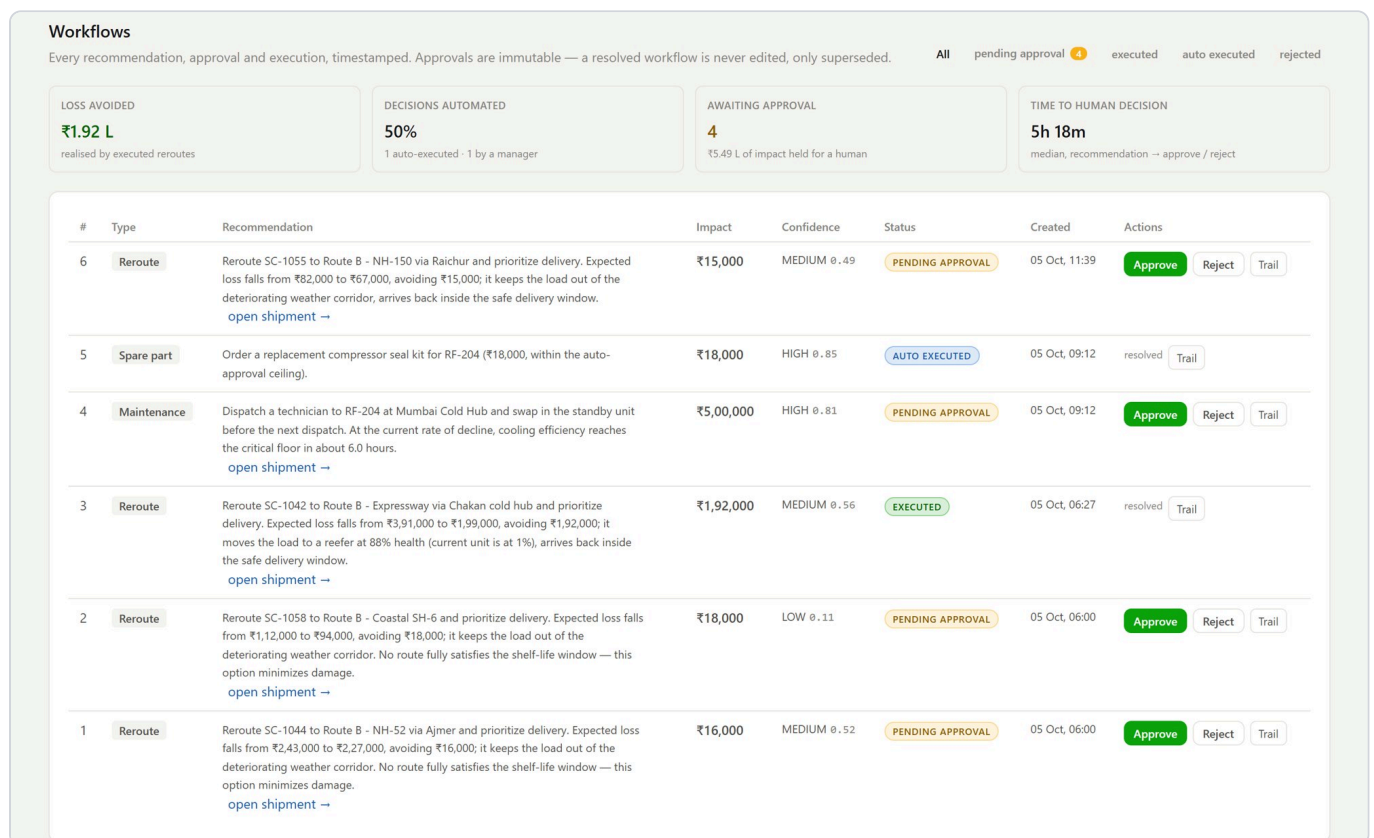


Figure 8. Workflows after the manager approves. Reroute #3 for SC-1042 is EXECUTED and books ₹1.92 L of loss avoided. A ₹18,000 spare-part order for RF-204 was AUTO EXECUTED (high confidence, under ₹25,000). Four more recommendations wait for a human. Approvals are timestamped and immutable.

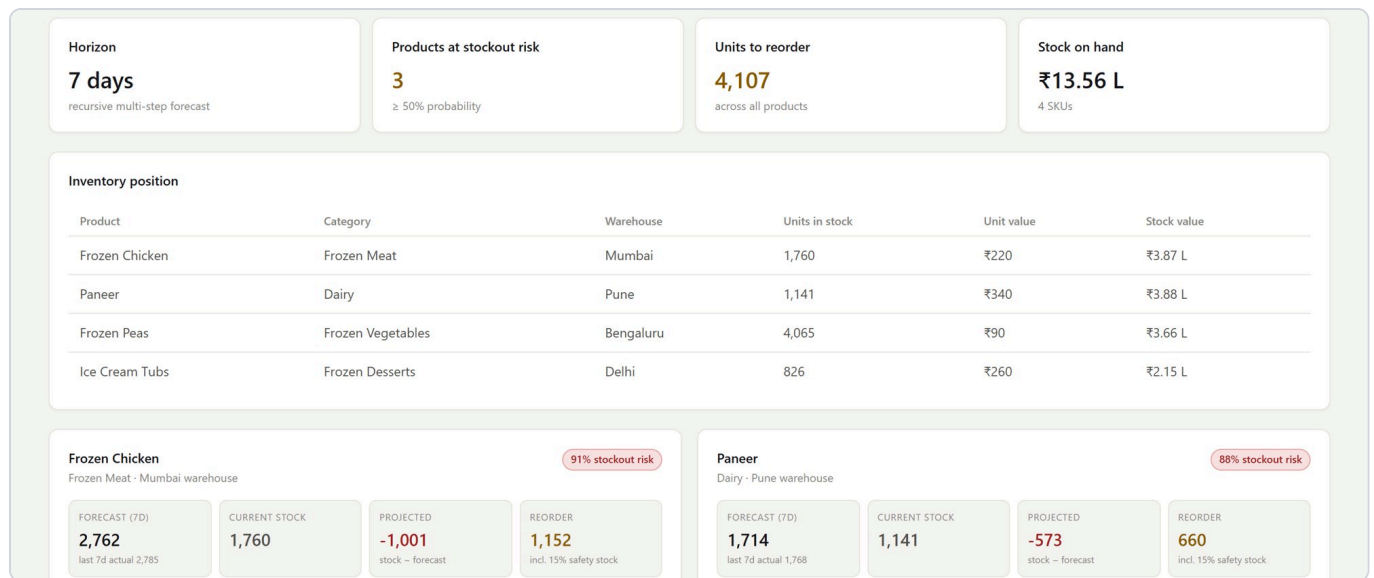


Figure 9. Supply-chain forecast. 7-day demand per product from the gradient-boosting models, set against stock on hand. Three products are at stockout risk; the system recommends reordering 4,107 units in total, including 15% safety stock. A load lost to spoilage shows up here as demand that still has to be met.

The demo, step by step

#	BUTTON PRESSED (A ROOT CAUSE)	WHAT THE SYSTEM WORKS OUT ON ITS OWN
1	Traffic incident on the lane	ETA margin shrinks and shelf life keeps burning. SAFE → LOW. Cargo stays at -18 °C, with no excursion.
2	Refrigerant leak on RF-204	Efficiency falls, sensors drift, the anomaly detector fires and failure risk reaches 91%. The <i>shipment</i> jumps to HIGH and takes the top slot.
3	Heat + storm front	Heat ingress beats the weakened cooling, the cold chain breaks and excursion minutes start to accrue. CRITICAL, ~77% spoilage.
4	Approve recommendation	The load moves to Route B and the standby reefer. Expected loss roughly halves and ₹1.92 L avoided lands in the ACT stage.
5	Repair RF-204	The wear rate returns to normal and the cargo cools again. Nobody lowered a score, and damage already done is not undone.

Built for a judge using it alone

- **► Guided demo** runs the five steps itself, jumping to the page where each consequence shows up.
- **Always moving.** The board is live on arrival: the clock advances, shelf life burns and congestion follows rush hour.
- **Bounded episodes.** The twin pauses when nobody is watching and resets for a visitor returning later.
- **Provenance line** on every page says which streams are live and which are simulated.
- **Offline-safe.** With no internet, weather and map fall back and say so; the demo still completes.
- **Open API.** Every screen is backed by a documented endpoint at /docs.

9 Business Impact

Measured on the live prototype, OpsGenie turns a load heading for a write-off into a priced, approved decision, and keeps humans in charge of every high-stakes action.

49%

cut in expected loss on the at-risk load (₹3.91 L → ₹1.99 L)

58%

lower forecast error, so smaller safety stock for the same service level

0.4%

false-alarm rate of the anomaly detector on healthy units

100%

of actions above ₹1 L approved by a human before they run

IMPACT AREA	MECHANISM	QUANTIFIED EFFECT
Spoilage reduction	Joins reefer health, weather and traffic into one risk score; reroutes to a healthy unit before the breach	₹1.92 L avoided on one ₹5 L load in the demo scenario
Maintenance cost	Failure-in-24 h prediction replaces calendar service and breakdown repair	ROC-AUC 0.987 on observable windows; flags fast leaks hours before the floor
Inventory & working capital	ML demand forecast and reorder quantity per product	55–63% lower error than a moving average; stockouts flagged 7 days ahead
Manager time	One ranked priority card instead of a wall of alerts; low-impact actions run on their own	Small, confident actions (e.g. an ₹18K spare-part order) auto-execute
Risk & governance	Confidence × impact gate, materiality floor on reroutes, immutable audit log	No high-impact action without a timestamped human approval
Customer experience	Loads arrive in spec and stock is reordered before it runs out	Fewer rejected deliveries and empty shelves for retail and pharma clients

Illustrative distributor scenario

Assume a cold-chain distributor writes off **₹1 crore a year** to spoilage. The measured 49% cut in expected loss is applied to the share of spoilage events caught early enough to act. That share depends on how much of the fleet is sensor-equipped, so three levels are shown. **These are assumptions applied to one measured rate, not a forecast.**

EVENTS CAUGHT IN TIME	LOSS CUT PER CAUGHT EVENT	ANNUAL LOSS REMAINING	SAVING VS. NO SYSTEM
No control tower	—	₹1,00,00,000	—
25% (pilot, part of the fleet)	49%	₹87,75,000	₹12,25,000
50%	49%	₹75,50,000	₹24,50,000
75% (full fleet with sensors)	49%	₹63,25,000	₹36,75,000

On top of this come fewer emergency repairs, lower safety stock from a sharper forecast, and the manager hours saved by a ranked queue. These depend on each operator's fleet and are not added to the figures above.

Cost to run

CPU only, no GPU and no paid API: training takes about one minute on a laptop and live weather is free. The whole stack runs on one small server or free hosting tiers; scoring each asset is a handful of tree-model calls every few seconds, so the cost per shipment is negligible.

10 Feasibility

Implementation requirements

- Reefer telemetry (temperature, current, vibration, pressure) from existing IoT gateways or telematics.
- GPS and a traffic/routing API for live ETAs (plugs into the provider chain that already exists).
- Order and stock feed from the ERP / WMS for forecasting.
- Integration with courier, CMMS and ERP so approved workflows execute in real systems.

Resources

- Team of 4: ML, backend, frontend, logistics-domain lead.
- CPU-only compute; four models retrain in about a minute.
- One historical season of real fleet data to recalibrate the Melt Index weights and thermal constants.

Technical challenges and mitigations

Synthetic training data	Retrain on real fleet history; pipeline and sanity gates are data-agnostic
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Unvalidated weights	Fit Melt Index weights and thermal constants to real spoilage outcomes
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Sensor gaps	Per-item fallback with honest SIMULATED labels; the score never silently uses stale data
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Trust in automation	₹1 L hard ceiling, confidence gate, override and immutable audit trail
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Model drift	Monthly retraining; monitor AUC and forecast MAE against the baselines
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Scalability plan

SQLite → PostgreSQL is a one-line change. Move telemetry to a stream (MQTT / Kafka), push updates over websockets instead of polling, and run the physics and scoring workers horizontally per region.

11 Future Scope

HORIZON	ENHANCEMENTS
Next 3 months	Real IoT and traffic providers; PostgreSQL; authentication and roles; websocket push; SMS / WhatsApp alerts to drivers and managers
6–12 months	ERP / WMS / courier execution of approved workflows; survival-model RUL; multi-stop route optimization; pharma cold chain (2–8 °C vaccines)
12+ months	LLM copilot with function-calling into these APIs (the language layer never computes a risk score); per-client thresholds; a self-serve control tower for small distributors

12 Conclusion

OpsGenie AI shows that operations AI is most valuable when it ends in a **decision**, not a dashboard. It joins the four parts of the track (predictive maintenance, supply-chain forecasting, operations optimization and workflow automation) into one causal pipeline, where a failing machine visibly changes the risk, the route and the money on the shipment it carries.

Every score breaks down into its causes. Every action is priced in rupees. Small, confident actions run on their own, and big ones wait for a human. The failure model reaches 0.987 ROC-AUC, the forecast cuts error by 58%, and in the live scenario one approved reroute avoids ₹1.92 L of loss.

Try it: [open the app](#) → ► **Guided demo**, or press the scenario buttons in order: Traffic incident → Refrigerant leak → Heat + storm → **Approve** → Repair. Watch SC-1042 go from SAFE to CRITICAL and back to a priced, approved recovery.